Interpretable Regional Descriptors

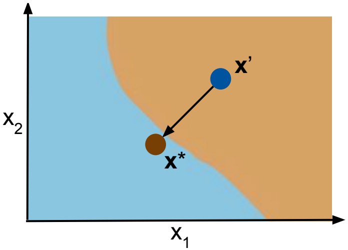
Susanne Dandl

30.04.2026

Counterfactual vs. Semi-factual Explanations

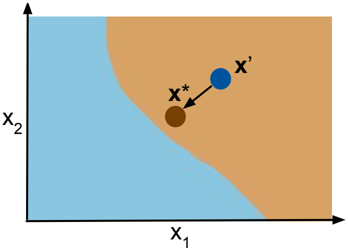
Counterfactual Explanations

Minimal changes in a few features required for changing a prediction



Semi-factual Explanations

Maximal changes in a few features required for **not** changing a prediction



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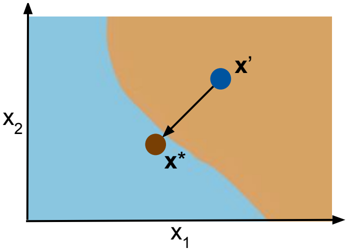

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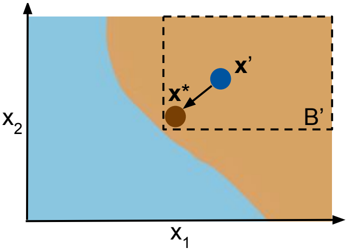

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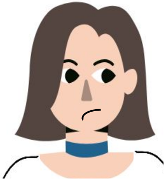
Motivation for Interpretable Regional Descriptors

How can the features be changed, such that the model still rates the credit as a moderate risk?



Feature	Customer	IRD	1-dim IRD
sex	female	{female, male}	{female, male}
saving.accounts	little	{little, moderate, rich}	{little, moderate, rich}
purpose	car	{car, radio/TV, furniture, others}	{car, radio/TV, furniture, others}
age	22	[19, 22]	[19, 75]
job	skilled	{skilled, highly skilled}	{unskilled, skilled, highly skilled}
housing	rent	{rent}	{own, free, rent}
checking.account	moderate	{little, moderate}	{little, moderate}
credit.amount	4000	[4000, 5389]	[2127, 8424]
duration	30	[26, 33]	[6, 44]

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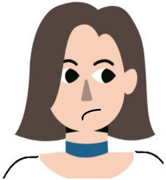

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1. Provide a set of semi-factual explanations to justify a prediction (“even-if” statements)
   “Even if you increase your savings and have a longer payback period,
   your credit would still be of moderate risk”

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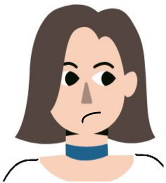

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2. Help with model auditing
“If male customers are not part of the IRD this can be an indicator for pointwise biases”

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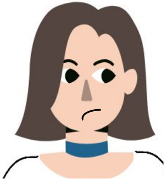

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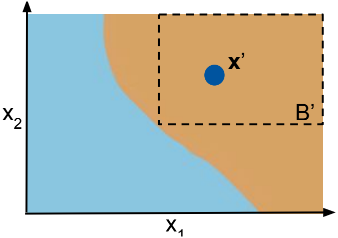
3. Identify whether a feature affects a prediction locally
“Compared to credit amount or duration, savings or purpose seem less relevant because their IRD encompasses the entire observed feature ranges”

General Task



Given: ML model $\hat{f} : \mathcal{X} \to \mathbb{R}$, point of interest $\mathbf{x}'$

Aim: Identify the largest hyperbox B' among all hyperboxes B which cover $\mathbf{x}'$ and only contain observations with predictions in $Y' = [\hat{f}(\mathbf{x}') - \epsilon_L, \hat{f}(\mathbf{x}') + \epsilon_H], \epsilon_L, \epsilon_H \in \mathbb{R}_{\geq 0}$

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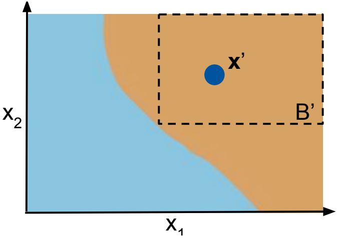

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Optimization problem:

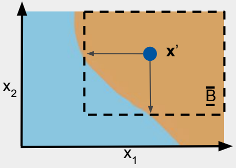
$$B' \in \underset{B}{arg\,max}(\underbrace{\mathbb{P}(\mathbf{x} \in B | \mathbf{x} \in \mathcal{X})}_{=:coverage(B)})$$

$$\text{s.t. } \underbrace{\mathbb{I}(\mathbf{x}' \in B)}_{=:locality(B)} = 1 \text{ and } \underbrace{\mathbb{P}(\hat{f}(\mathbf{x}) \in Y' | \mathbf{x} \in B)}_{=:precision(B)} = 1$$

General Framework - 4 Steps

1

Definition of search space



2

3

4

General Framework - 4 Steps

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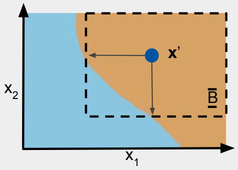

Theorem

For any B' solving the above optimization task:

$$B' \subseteq \overline{B}$$

2

3

4

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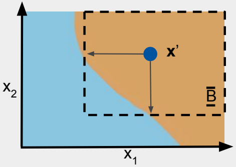

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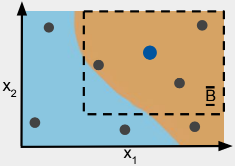
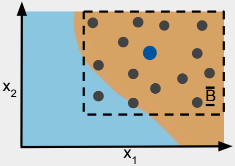
2 Selection of data set



training data



sampled data

3

4

General Framework - 4 Steps

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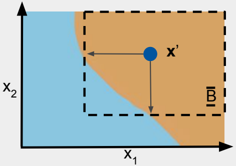

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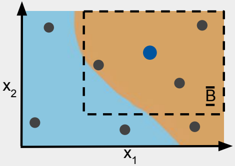

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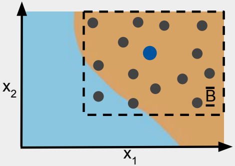

sampled data

3

4

RQ: What effect does sampled data have compared to training data?

General Framework - 4 Steps

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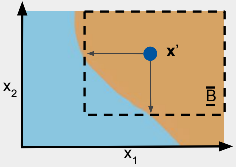

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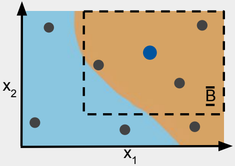

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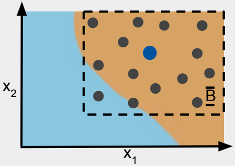
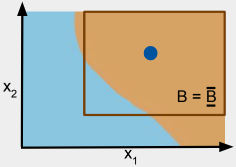
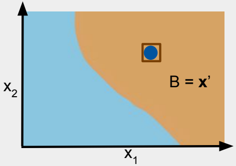


sampled data

3 Initialization of box



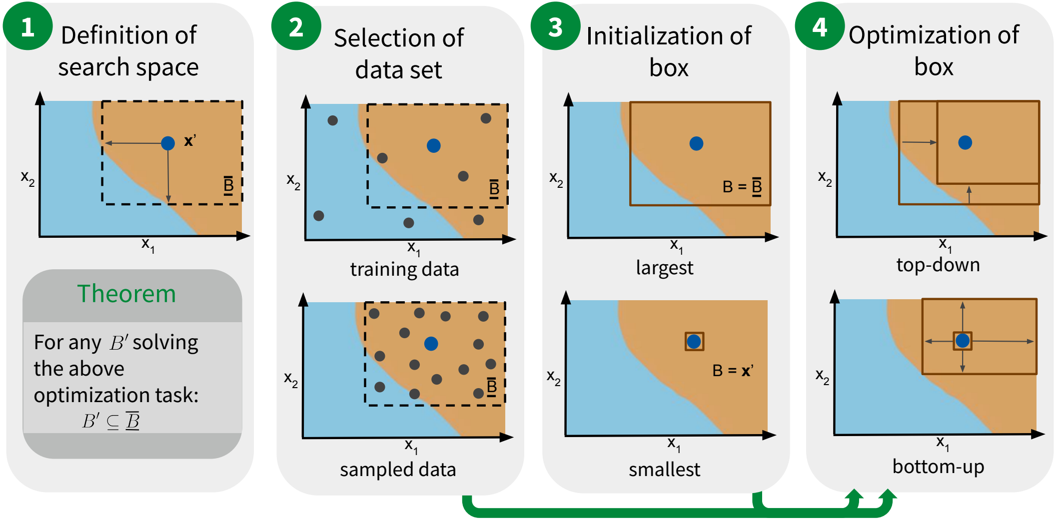
largest



smallest

4

General Framework - 4 Steps



Review of Hyperbox Methods

Desiderata:

- Interpretable: return a single hyperbox
- Model-agnostic: work for regression/classification models and various feature types
- Sparse: adhere to optional user-defined immutability constraints

Review of Hyperbox Methods

	Objectives			Desiderata		
	Coverage	Precision	Locality	Interpretable	Agnostic	Sparse
Level set methods						
PBnB (Zabinsky et al. (2011, 2020))	$\surd$	$\surd$	$\times$	$\times$	$\surd$	$\times$
Data mining methods						
MaxBox (Eckstein (2002))	$\surd$	$\surd$	$\times$	$\surd$	$\times$	$\times$
PRIM (Friedman & Fisher (1999))	$\times$	$\times$	$\times$	$\surd$	$\times$	$\times$
Local interpretation methods						
Anchors (Ribeiro et al. (2018))	$\surd$	$\surd$	$\surd$	$\surd$	$\times$	$\times$
MAIRE (Sharma et al. (2021))	$\surd$	$\surd$	$\surd$	$\surd$	$\times$	$\times$
LORE (Guidotti et al. (2019, 2022))	$\times$	$\times$	$\surd$	$\surd$	$\surd$	$\times$
Interpretable classifiers						
Column generation (Dash et al. (2018))	$\surd$	$\surd$	$\times$	$\times$	$\surd$	$\times$

Review of Hyperbox Methods

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	Coverage	Precision	Locality	Interpretable	Agnostic	Sparse
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MaxBox (Eckstein (2002))	$\surd$	$\surd$	$\times$	$\surd$	$\times$	$\times$
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Local interpretation methods						
MAIRE (Sharma et al. (2021))	$\surd$	$\surd$	$\surd$	$\surd$	$\times$	$\times$

Baseline: Anchors (Ribeiro et al. (2018))

Eckstein et al. (2002). *The Maximum Box Problem and its Application to Data Analysis*. Computational Optimization and Applications.
Friedman, Fisher (1999). *Bump Hunting in High-dimensional Data*. Statistics and Computing.
Sharma et al. (2021). *MAIRE - A Model-agnostic Interpretable Rule Extraction Procedure for Explaining Classifiers*. Machine Learning and Knowledge Extraction.
Ribeiro et al. (2018). *Anchors: High-precision Model-agnostic Explanations*. AAAI.

Review of Hyperbox Methods

	Objectives			Desiderata		
	Coverage	Precision	Locality	Interpretable	Agnostic	Sparse
Data mining methods						
MaxBox (Eckstein (2002))	√	√	√	√	√	√
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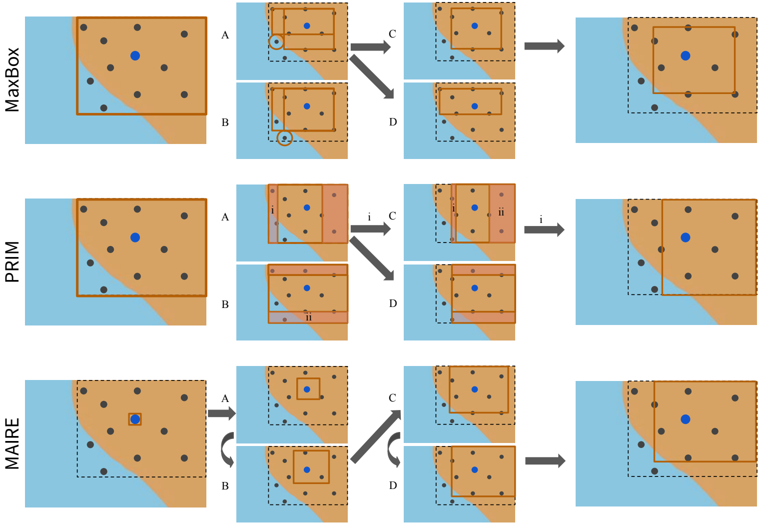
MAIRE (Sharma et al. (2021))	√	√	√	√	√	√

Baseline: Anchors (Ribeiro et al. (2018))

RQ: How do MaxBox, PRIM and MAIRE perform against each other?

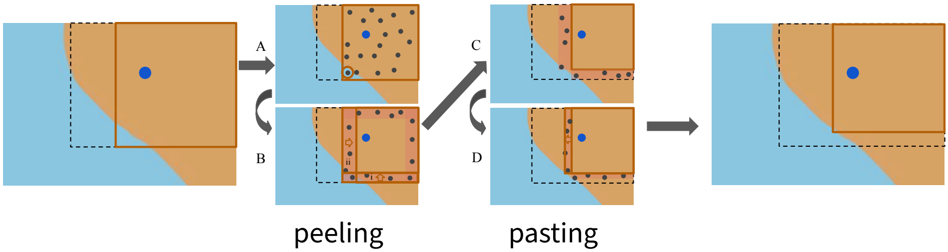
Review of Hyperbox Methods

Input: initial box & data set



Post-Processing

Input: generated box & data sampler



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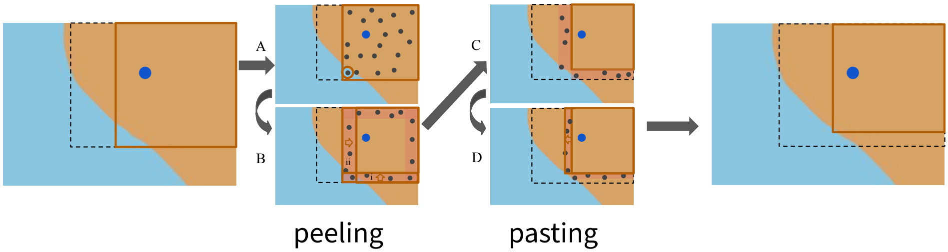

RQ: What effect does post-processing have?

Research Questions

1. How do MaxBox, PRIM and MAIRE perform against each other?
2. What effect does (double-in-size) sampled data have compared to training data as the underlying dataset?
3. What effect does post-processing have?

Quality Measures

- **Locality**: Does the box B' contain the data point $\mathbf{x}'$ we are interested in? $locality(B)$
- **Coverage**: How much of the feature space is covered by B'? $\widehat{coverage}(B)$
- **Precision**: Do all data points $\in B'$ have the same prediction as $\mathbf{x}'$? $\widehat{precision}(B)$
- **Efficiency**: How often is $\hat{f}$ called by a method?

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- **Precision**: Do all data points $\in B'$ have the same prediction as $\mathbf{x}'$? $\widehat{precision}(B)$
- **Efficiency**: How often is $\hat{f}$ called by a method?
- **Maximality**: Is the box maximal, so that no more boxes can be added under full precision?
- **Robustness**: Does an IRD method ran multiple times on the same data point and model, will produce B's that overlap?

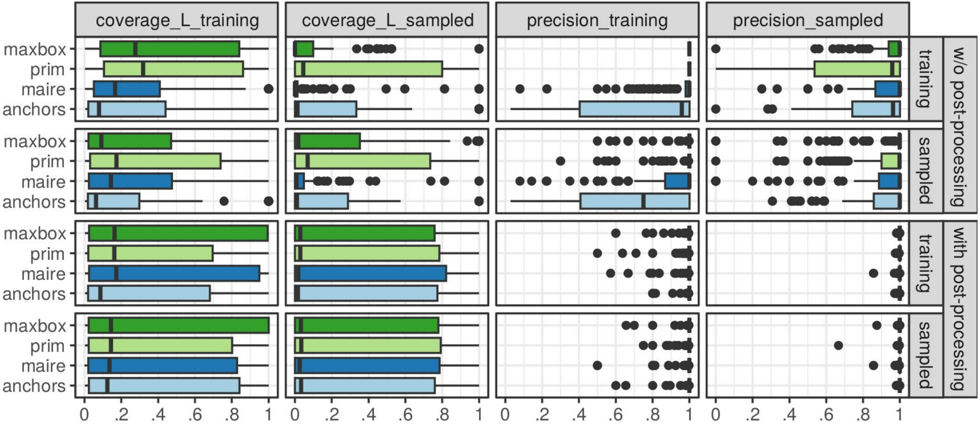
Setup

- 6 datasets

Name	ID	Type	n	Cont.	Categ.
diabetes	37	binary	768	8	0
tic_tac_toe	50	binary	958	0	9
cmc	23	three-class	1473	2	7
vehicle	54	four-class	846	18	0
no2	886	regression	500	7	0
plasma_retinol	511	regression	315	10	3

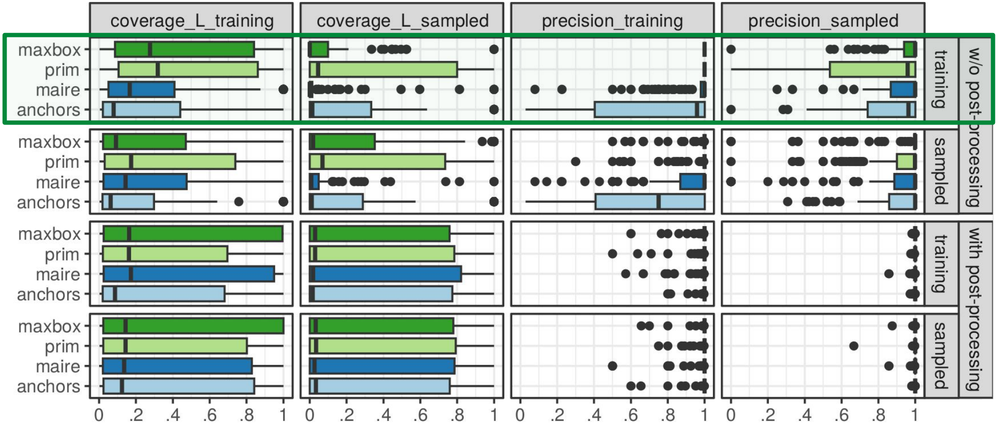
- Models: hyperbox, logistic/multinomial/linear, neural network, random forest
- Methods: MaxBox, PRIM, MAIRE, Anchors (baseline), either with or without post-processing
- Underlying data: training data or double-in-size newly sampled data points within $\bar{\underline{B}}$

Results



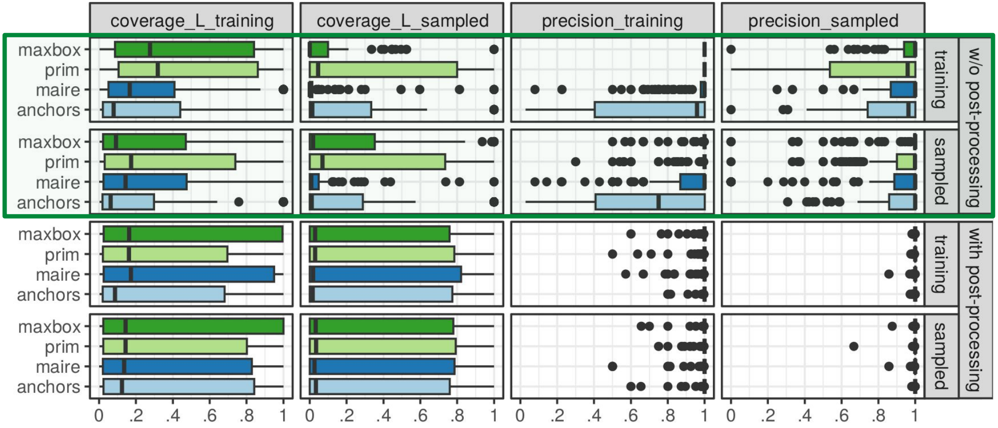
coverage_L_training
coverage_L_sampled
precision_training
precision_sampled
maxbox
prim
maire
anchors
training
sampled
w/o post-processing
with post-processing
0
.2
.4
.6
.8
1

Results - RQ 1) Methods



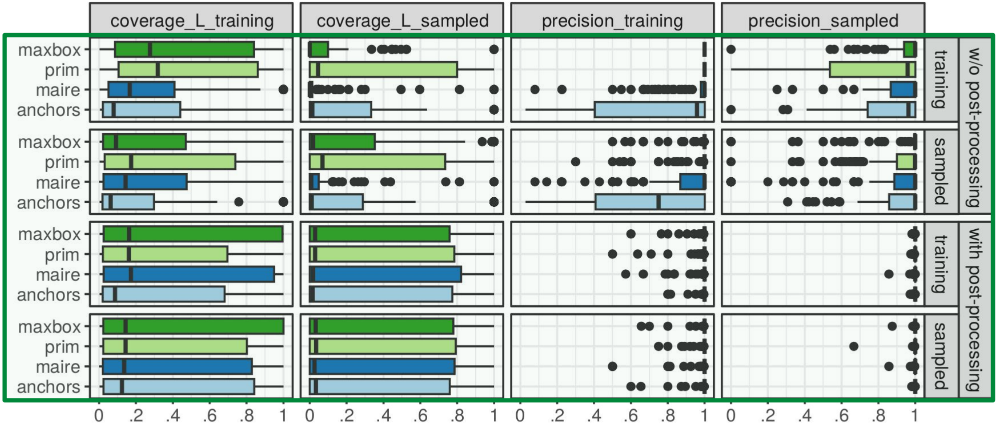
RQ 1) All fulfill locality, PRIM highest coverage, MaxBox highest precision, equal no. calls to $\hat{f}$

Results - RQ 2) Underlying Data Set



RQ 2) Double-in-size sampled data higher coverage, precision w.r.t. sampled data

Results - RQ 3) Post-processing



RQ 3) Post-processed boxes higher coverage and precision, but also higher no. of calls to $\hat{f}$

Summary & Open Issues

Summary of contributions:

1. New form of explanation for single predictions
2. Embedment of three existing hyperbox-based methods in a unifying framework
3. Identification of strategies to further enhance quality (post-processing and sampling)

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Open Issues:

- Faithfulness to the DGP and not only to the model
- Strategies to restrict the search space for high-dimensional data
- Sensitivity analysis w.r.t. to methods' hyperparameters, slight changes in $\mathbf{x}'$, etc.
- Application of IRDs to other use cases